5. What is the principal start codon and to what amino acid does it correspond?
6. Restriction endonucleases are naturally found in bacteria. What purpose do they serve?
7. There are many varieties of cloning vectors that are used to propagate cloned DNA. One type of cloning vector used in E.coli is a plasmid vector. What features does a plasmid vector have that makes it useful for constructing and cloning recombinant DNA molecules?
8. What is shuttle vector and why is it used?
9. What information and materials are needed to amplify a segment of DNA using PCR?
10. In most PCR reactions, a DNA polymerase that can withstand short periods of very high (near boiling) temperatures is used. why?
11. The sequence of nucleotides in an mRNA is 5′-AUG-ACCCAUU-CAUUGGUCUCGUUAG-3′. Assuming that ribosomes could translate this mRNA, how many amino acids long would you expect the resulting polypeptide chain to be?
12. The N-terminus of a protein has the sequence Met-His-Arg-Lys-Val-His-Cys-Gly. A molecular Biologist wants to synthesize a DNA chain that can encode this portion of the protein. How many DNA sequences can encode this polypeptide?
13. Explain the process by which an infected bacterium releases progeny phage.
14. Define coding strand.
15. Distinguish a missense and a nonsense mutation.
16. By what mechanism does 5-bromouracil induce mutations?
17. Define the term conjugative.
18. How does an Hfr cell differ from F^+ cell?
19. How are F′ plasmids produced?
20. Define a lysogen.
21. Restriction enzymes generate two types of termini. What are they?
22. Explain cosmids and the advantages resulting from the use of a cosmid?
23. Explain the use of bacteriophage in cloning DNA fragment.
24. What are expression vectors?
25. Diagramatically describe the plasmid cloning vector PUC19.
26. How is the natural phenomenon of conjugation used to transfer foreign gene?
27. List the stages involved in Recombinant DNA technology.
28. Discuss RAPD and RFLP.

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